CORTEX — Clinical Validation Module

Module Specification v1.1 — February 2026

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1. Overview

The Validation module bridges the gap between "what the SOA says we need to achieve" (benchmarks from the SOA Device module) and "what our device actually achieves" (test results from the Data Science team). It manages the full lifecycle: protocol design, sample size justification, data import, results mapping, and report generation.

Two distinct study types, each with its own protocol and report structure:

Type	Purpose	Design	Key Output
Standalone	Demonstrate software performance against ground truth	Device vs. Ground Truth (confusion matrix)	Validation Report (performance)
MRMC	Demonstrate clinical benefit with radiologist readers	Multi-Reader Multi-Case, Aided vs. Unaided arms	Clinical Benefit Report

A project may have multiple validation studies: an initial standalone, an MRMC for clinical benefit, and subsequent standalone re-runs for software patches. Each study has its own protocol and results, but they all reference the same SOA benchmarks.

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2. Upstream Dependencies

```
SOA Clinical          SOA Device            SLS (optional)
    │                   │                   │
    │ endpoints that matter   │ performance benchmarks   │ radiologist
    │ clinical context        │ similar device perf.     │ performance
    │                   │ acceptance criteria      │ literature
    │                   │ quality assessments      │
    └─────────────────┬─────────────────┘                   │
          │                         │
          ▼                         │
    ┌──────────────────────────┐                   │
    │  VALIDATION MODULE  │◀──────────────────────────────┘
    │                 │
    │ Protocol (Study     │
    │ Summary)            │
```

```
│      │          │
│      ▼          │
│ Data Import (XLS)   │
│      │          │
│      ▼          │
│ Validation Report   │──────▶ CER Module
│                 │──────▶ SOA Comparison Table
└──────────────────────────────┘
```

Key upstream links:

- SOA benchmarks define the performance targets and acceptance criteria
- SOA similar device data provides the comparator in the report
- SOA quality assessments (QUADAS-2 + Reading Grids) inform the evidence quality context referenced in the validation discussion
- An optional mini SLS can be conducted to find radiologist performance benchmarks for MRMC study design

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3. Data Model

3.1 Hierarchy

```
Project
└─ Validation Study
   ├─ Study Configuration (type, version, linked SOA)
   ├─ Protocol (Study Summary)
   │  ├─ Endpoints (Primary, Secondary, Additional Assessments)
   │  ├─ Dataset Description
   │  ├─ Readers (MRMC only)
   │  ├─ Statistical Strategy
   │  └─ Sample Size Justification
   ├─ Data Imports (versioned XLS deliveries from Data Science)
   │  ├─ Data Distribution / Stratification
   │  └─ Statistical Results (per endpoint)
   ├─ Results Mapping (endpoint → result → benchmark comparison)
   └─ Generated Report (DOCX)
```

3.2 Validation Study Record

Field	Description
id	UUID

Field	Description
project_id	Parent project
name	e.g., "CINA-CSpine v1.0 Standalone Validation"
study_type	standalone | mrmc | standalone_patch
version	Study version (e.g., "1.0")
status	draft → protocol_approved → data_collection → analysis → report_draft → completed → locked
linked_soa_device_id	SOA Device analysis providing benchmarks
linked_soa_clinical_id	SOA Clinical analysis providing context
parent_study_id	For patches: reference to the original standalone study (nullable)
software_version	Device software version being validated
regulatory_context	CE-MDR | 510k | De Novo | NMPA
notes	Free-form

4. Protocol (Study Summary)

The protocol is the regulatory document defining what will be tested, how, and what success looks like. It must be finalized before testing begins — a Notified Body will verify that acceptance criteria were not defined after seeing results.

4.1 Protocol Sections

The protocol follows a structured template matching your current document format:

```
1. Introduction
  ├── 1.1 Document Overview
  ├── 1.2 Abbreviations (auto-managed)
  ├── 1.3 Project References (auto-linked to other CORTEX modules)
  └── 1.4 Intended Use (pulled from project definition)

2. Methodology
  ├── 2.1 Material (device description, software version)
  ├── 2.2 Evaluated Data
  │   ├── 2.2.1 Data Collection Process
  │   ├── 2.2.2 Inclusion Criteria
  │   └── 2.2.3 Exclusion Criteria
  ├── 2.3 Data Assessment
  │   ├── 2.3.1 Ground Truth Process (standalone)
  │   │   OR Reader Qualification & Training (MRMC)
  │   ├── 2.3.2 Final Consensus Database
  │   └── 2.3.3 Post-Processing & Results Reporting
  └── 2.4 Stratification Plan

3. Statistical Analysis
  ├── 3.1 Primary Endpoint
  │   ├── Definition & Performance Goal
  │   └── Sample Size & Targeted Performance Estimation
  ├── 3.2 Secondary Endpoint
  ├── 3.3 Additional Supportive Statistics
  └── 3.4 Additional Assessments

4. Acceptance Criteria Summary
```

4.2 Endpoints

Each study defines a structured set of endpoints across three tiers:

Endpoint Record:

Field	Description
id	UUID
study_id	Parent study
tier	primary | secondary | additional_assessment
name	e.g., "Sensitivity", "Interpretation Time", "Fracture Localization PPV"
description	Detailed definition of the endpoint
metric_type	percentage | time | count | score | agreement | p_value
statistical_method	e.g., "Clopper-Pearson exact binomial", "Paired t-test", "Cohen's Kappa"
linked_benchmark_id	SOA Device benchmark this endpoint maps to (nullable — not all endpoints have an SOA benchmark)
acceptance_criterion	e.g., "≥ 60%", "p < 0.05", "≥ 85%"
acceptance_justification	Why this criterion was chosen (required — regulatory traceability)
ci_method	e.g., "Clopper-Pearson 95% CI", "Normal approximation"
ci_width_target	Target CI width (e.g., "± 10%")
sort_order	Display order

Default Endpoints by Study Type:

Standalone:

Tier	Endpoint	Typical Method	Typical Acceptance
Primary	Sensitivity	Exact binomial, Clopper-Pearson 95% CI	≥ SOA lower range
Primary	Specificity	Exact binomial, Clopper-Pearson 95% CI	≥ SOA lower range
Secondary	Processing Time	Descriptive (Mean ± SD, Median, Min, Max)	Comparable to similar device
Secondary	Time-to-notification (TP only)	Descriptive	Comparable to similar device
Additional	Accuracy	(TP+TN)/(TP+FN+FP+TN), 95% CI	Informative
Additional	PPV / NPV	At varying prevalence (5-50%, increments of 5%)	Informative
Additional	AUC (ROC)	From sensitivity/specificity	Informative
Additional	MCC	Matthews Correlation Coefficient	Informative
Additional	Localization PPV	TP BBs / (TP + FP) BBs	Informative
Additional	Vertebral Labeling Agreement	% overall agreement	Informative

MRMC (covers both simple CSpine-like and complex ASPECTS-like designs):

Tier	Endpoint	Typical Method	Typical Acceptance
Primary	Assessment Accuracy (% regions matching GT)	Bootstrap 95% CI, paired difference Aided vs. Unaided	Statistically significant improvement ($p < 0.05$)
Primary	Interpretation Time (per-case)	Paired comparison, Aided vs. Unaided	Statistically significant reduction ($p < 0.05$)
Secondary	Reading Time (per-case)	Paired comparison, Mean ± SD, 95% CI	Statistically significant reduction
Secondary	Scan-to-Assessment Time	Paired comparison	Assessed only if primary is met
Additional	ROC AUC (ordinal scale)	Obuchowski-Rockette ANOVA + Jackknife	Informative — AUC difference Aided vs. Unaided
Additional	Stratified accuracy by reader expertise	Bootstrap 95% CI per subgroup (expert vs. non-expert)	Informative
Additional	Stratified accuracy by region/anatomy	Bootstrap 95% CI per region group (e.g., deep vs. cortical)	Informative
Additional	Per-reader analysis	Individual reader results (accuracy, time)	Informative
Additional	Region-based Sensitivity/Specificity/PPV/NPV	Binary analysis using rating scale cutoff	Informative
Additional	Score-based inter-reader reliability	ICC, Cohen's/Fleiss' Kappa	Informative
Additional	Score-based correlation vs. GT	Linear regression, Pearson/Spearman correlation	Informative

Endpoints are editable — the user can add, remove, rename, change methods. But once the protocol status moves to `protocol_approved`, changes trigger a warning (see Section 9.2).

4.3 Dataset Description

Dataset Record:

Field	Description
study_id	Parent study
description	Narrative description of the dataset
clinical_sources	List of institutions providing data (name, country, type)
acquisition_periods	Per source: date ranges of data acquisition
total_cases_target	Target N from sample size calculation
inclusion_criteria	Structured list (matches your current format)
exclusion_criteria	Structured list
positive_negative_balance	Target distribution
separation_from_training	Statement confirming data separation from training set

Stratification Plan:

Define the dimensions along which data distribution will be reported and (optionally) stratified analysis performed:

Dimension	Categories	Type
Scanner maker	GE, Philips, Siemens, Canon/Toshiba, Fujifilm, Other	Imaging
Scanner model	Per maker (count)	Imaging
Number of detector rows	< 64, ≥ 64	Imaging
Slice thickness	< 1mm, 1-1.25mm, > 1.25mm	Imaging
Reconstruction kernel	Bone/Sharp, Soft tissue	Imaging
Clinical source	Per institution	Dataset
Acquisition period	Per source	Dataset
Patient age	18-45, 45-75, ≥75 (configurable ranges)	Demographics
Patient sex	Male, Female, NA	Demographics
Case status	Positive, Negative	Outcome

Stratification dimensions are configurable — the user can add, remove, rename categories. This plan is defined in the protocol and must match the data distribution tables in the results.

4.4 Readers (MRMC Studies Only)

Reader Record:

Field	Description
study_id	Parent study
abbreviation	e.g., "JS", "PW"
full_name	Dr. Jennifer Soun
affiliation	University California Irvine
specialty	Neuroradiology
expertise_level	expert | non_expert
years_experience	14
board_certification	Board-certified neuroradiologist
training_completed	Boolean — did they complete training + proficiency test
notes	Additional qualifications

Study design parameters (MRMC):

Parameter	Description	Example (CSpine)	Example (ASPECTS)
study_design	Study design description	Retrospective, multi-center, concurrent, fully-crossed MRMC	Retrospective, concurrent, cross-over, fully-crossed MRMC
n_readers	Total readers	4	8
n_experts	Expert readers	2 (neuroradiologists)	4 (neuroradiologists)
n_non_experts	Non-expert readers	2 (residents)	4 (neurologist, general radiologist, stroke neurologist, neurointensivist)
n_cases	Cases per arm	100	200
arms	Study arms	["Unaided (Pre-AI)", "Aided (Post-AI)"]	["Unaided", "Aided"]
arm_order	How arms are sequenced	sequential (Unaided first, then Aided)	randomized_crossover (randomized per reader)
washout_period	Time between arms	1 month	4 weeks
rating_scale	How readers record assessments	binary (positive/negative)	ordinal_6point (Definitely/Probably/Possibly × Not Infarcted/Infarcted)
analysis_unit	Unit of analysis	per_case	per_region (10 ASPECTS regions per case)
blinding	Blinding description	Readers blinded to GT and each other	Readers blinded to GT, each other, and own prior assessments (locked after validation)
randomization	Randomization approach	None (fixed order)	Separate randomization scheme per reader (uniform distribution) for case order and arm order
training_protocol	Reader training description	Remote training + proficiency test	Remote training + proficiency test + practice cases (6 different files)
proficiency_test	Test description	Short test on platform use	Test on platform use, device outputs, and rating scale
platform	Evaluation platform	Dedicated reading platform	Dedicated platform with locked submissions, no re-review allowed

4.5 Statistical Strategy Assistant & Sample Size Calculator

The statistical strategy assistant helps the user design the study's statistical approach and calculate the required sample size. It operates in three interconnected modes.

Mode 1 — Forward Calculator (Classical)

"I know what I want to prove. How many cases do I need?"

Inputs:

- Target performance (from SOA benchmarks): e.g., sensitivity ≥ 60%, specificity ≥ 89%
- Desired CI width: e.g., ± 10% (Clopper-Pearson) or ± 5% (Bootstrap)
- Power: typically 80% ($\beta = 0.2$)
- Significance level: typically $\alpha = 0.05$
- Expected prevalence: e.g., 50% (CSpine) or 30% (ASPECTS regions)
- Analysis unit: per_case (CSpine) or per_region (ASPECTS: 10 regions/case)
- CI method: clopper_pearson | bootstrap | normal_approximation

Outputs:

- Minimum required N (at analysis unit level)
- If per-region: minimum cases = N_regions / regions_per_case
- Required positive and negative cases
- Calculation method reference (Hajian-Tilaki)

Example — CSpine:

> Target: Sensitivity ≥ 60%, Specificity ≥ 89%, CI ± 10%, Power 80%, Prevalence 50%
> → Minimum 200 cases (100 positive, 100 negative)

Example — ASPECTS:

> Target: Sensitivity ≥ 65%, Specificity ≥ 80%, CI ± 5%, Power 80%, Prevalence 30%, 10 regions/case
> → Minimum 1,224 regions → 125 cases (≥ 366 positive regions)

Mode 2 — Reverse Calculator

"I have a fixed dataset. What can I statistically prove, and what methods should I use?"

Inputs:

- Available N (total cases, or positive/negative breakdown)
- Desired target performance (from SOA)
- Analysis unit: per_case or per_region
- Constraints (e.g., "only 2 scanner manufacturers", "70% from one clinical source")

Outputs — a ranked table of options:

Statistical Method	Required N	Achievable CI Width	Achievable Power	Feasible?
Clopper-Pearson exact binomial (± 10%)	200	± 10%	80%	✅ Yes
Clopper-Pearson exact binomial (± 5%)	750	± 5%	80%	❌ Need 750, have 200
Bootstrap CI (± 8%)	250	± 8%	80%	⚠️ Close (need 250, have 200)
Wilson score interval (± 10%)	180	± 10%	80%	✅ Yes
Fisher's exact test (superiority)	150	N/A	75%	✅ Yes (lower power)

Key feature: If the initial method requires more data than available, the system suggests alternative methods that work with the available N. Each alternative shows the trade-off (wider CI, lower power, different assumptions). The user can then make an informed decision.

Additional suggestions:

- "With N=200, stratified analysis by scanner maker (4 groups of ~50) will have wide CIs per stratum — consider grouping into 2 categories"
- "Your positive/negative ratio of 30/70 means sensitivity CI will be wider than specificity CI"

Mode 3 — AI-Assisted Design

"Help me design the best study given my constraints."

The user describes their situation in natural language:

> "We have access to about 300 brain NCCTs from GVI (Buffalo) and UCI (Irvine), acquired on GE, Siemens and Philips scanners. Expected prevalence of positive ASPECTS regions is about 30%. We want to test CINA-ASPECTS region-based sensitivity and specificity. The SOA says readers achieve about 65% sensitivity and 80% specificity. We also want to do an MRMC with 6-8 readers."

The LLM responds with a structured recommendation:

> 1. **Standalone design:**
> - Analysis unit: per-region (10 regions/case → 3,000 regions from 300 cases)
> - Sample size assessment: 3,000 regions >> 1,224 minimum → ample power
> - Recommended method: Bootstrap 95% CI for region-based (accounts for within-case correlation)
> - Target: Sensitivity ≥ 65% [lower CI ≥ 60%], Specificity ≥ 80% [lower CI ≥ 75%]

> - Stratification feasible: scanner maker (3 groups of ~100), age, sex ✅
>
> 2. **MRMC design:**
> - Recommended: cross-over, fully-crossed, randomized order
> - 8 readers (4 expert neuroradiologists, 4 non-expert) — aligns with 21 CFR §892.2060
> - Washout: 4 weeks minimum
> - Primary endpoint: % regions matching GT (Aided vs. Unaided)
> - ROC analysis: Obuchowski-Rockette ANOVA + Jackknife (MRMCaov R package)
> - Secondary: reading time (paired comparison)
> 3. **Limitations to acknowledge:**
> - Only 2 clinical sources (geographic bias)
> - No Canon scanners — consider documenting as known gap for PMS
> - Cross-over design assumes no learning effect beyond washout period

The AI recommendation is editable — the user can accept, modify, or reject each suggestion. Accepted elements auto-populate the protocol fields.

Sample Size Justification (Protocol Section)

Regardless of which mode(s) the user employed, the system auto-generates the sample size justification paragraph for the protocol. This paragraph follows the format from your existing documents:

> "In order to define the target performance of [device], Avicenna.AI took into account the information reported in the published scientific literature pertaining to the performance of the selected similar device: [similar device] ([manufacturer]). [SOA reference]. Overall, the similar device's performances, such as sensitivities ranged from X% to Y%. Based on this information, Avicenna.AI decided to target a Sensitivity of at least Z% and a Specificity of at least W% for the performance analysis. Subsequently, the sample size was calculated using [method], following [reference]. Setting an expected Sensitivity to Z% and a Specificity to W%, with a 95% [CI method] not wider than ±N%, a power of P%, and a prevalence of Q%, yield a minimum of N cases, with at least N_pos positive cases."

The user can edit this generated text.

4.6 Mini Literature Search (MRMC Methodology)

When designing an MRMC study, the team may need to find published benchmarks for radiologist performance (inter-reader agreement, reader sensitivity/specificity without AI) to justify the study design and expected effect size.

Implementation: The user can create a quick SLS session (type: `ad_hoc`) directly from the Validation module, specifically scoped to "radiologist performance in [pathology]." The SLS follows the same workflow as the main SLS module (query → fetch → screen → include), including the updated features:

- Exclusion reason codes (short codes for PRISMA)
- Live-calculated PRISMA flowchart with per-query, per-database breakdown

- AI relevance scoring with code-mapped exclusion suggestions

This mini-SLS is lightweight — no full SOA analysis needed. The included articles inform the MRMC statistical design.

The SLS session is linked to the validation study for traceability.

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5. Data Import (XLS from Data Science)

5.1 Import Workflow

The Data Science team runs the actual validation in R/Python and produces Excel files. These are imported into CORTEX.

Import flow:

1. User uploads an XLS/XLSX file
2. System detects the file structure (sheets, columns, data types)
3. Mapping step: User maps XLS content to CORTEX data structures:
    - Which sheet contains data distribution tables?
    - Which sheet contains statistical results?
    - Which columns map to which endpoints?
    - Which columns define stratification groups?
4. Mapped data is imported and stored
5. Previous import version is archived (not deleted)

5.2 Versioning

- Each study can have multiple data import versions (>5 is common)
- Each version is timestamped and numbered (v1, v2, v3...)
- The system keeps all versions but marks the latest as active
- The user can switch the active version (e.g., roll back to v3 if v4 had issues)
- The validation report always generates from the active version
- Version comparison: user can diff two versions to see what changed (N cases, result deltas)

5.3 Standardized Import Templates

To reduce the mapping burden, CORTEX provides import templates (downloadable XLS):

Standalone Template:

Sheet	Content
Metadata	Study name, software version, import date, data science contact
Cohort	Total cases, positives, negatives, exclusions with reasons
Data Distribution	One table per stratification dimension (scanner maker, age, sex, clinical source, etc.) with columns: Category, Count, Percentage
Confusion Matrix	Overall: TP, FN, FP, TN. Per stratum (optional): same.
Primary Endpoint	Metric, Value, CI Lower, CI Upper, N
Secondary Endpoint	Metric, Value, CI Lower, CI Upper, Median, Min, Max, SD
Additional Statistics	Metric, Value, CI Lower, CI Upper, N, Parameters (e.g., prevalence for PPV/NPV)
Additional Assessments	Metric, Value, N, Notes
Stratified Results	Stratum, Metric, Value, CI Lower, CI Upper, TP, FN, FP, TN

MRMC Template:

Sheet	Content
Metadata	Study name, software version, import date
Cohort	Total cases per arm, readers, exclusions
Data Distribution	Same as standalone
Reader Results	Reader, Arm, Metric, Value, SD
Overall Results	Arm, Metric, Mean, SD, CI Lower, CI Upper, Median, Min, Max
Paired Comparison	Metric, Unaided Mean, Aided Mean, Difference, p-value, CI
Stratified Results	Stratum (reader expertise / case type), Arm, Metric, Value

If the Data Science team uses the template, import is automatic (no mapping step). If they use a custom format, the user does the mapping manually.

5.4 Data Distribution Tables

Imported from XLS and displayed in CORTEX. These tables show the representativeness of the dataset.

For each stratification dimension:

- Category name
- Count (N)
- Percentage (%)
- Positive / Negative breakdown (optional)

Validation check: CORTEX verifies that the stratification dimensions match those defined in the protocol's stratification plan. Mismatch → warning.

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6. Results Mapping & Analysis

6.1 Endpoint → Result Mapping

Once data is imported, each endpoint defined in the protocol is linked to a result from the import:

Protocol Endpoint	Imported Result	SOA Benchmark	Status
Sensitivity (primary)	90.3% [84.5–94.5%]	≥ 60% (SOA range: 54.9–91.7%)	✅ PASS
Specificity (primary)	91.9% [86.8–95.5%]	≥ 89% (SOA range: 88.6–98.6%)	✅ PASS
Processing Time (secondary)	Mean: 12.3s ± 4.1s	Similar device: ~15s	✅ Comparable
Accuracy (additional)	91.2% [87.7–93.9%]	—	ℹ️ Informative
...	...	...	...

6.2 Pass/Fail Logic

For each endpoint:

- **PASS:** result meets the acceptance criterion defined in the protocol
- **FAIL:** result does not meet the criterion
- **INFORMATIVE:** no acceptance criterion (additional assessments)

When a result FAILS:

- The endpoint is flagged red in the UI
- A justification field becomes required — the user must explain why the result is acceptable despite not meeting the criterion, or document corrective actions
- The justification is included in the generated report's Discussion section
- Common justifications: CI overlap with benchmark, conservative criterion, known data limitation, subgroup analysis shows pass in key populations

6.3 SOA Comparison

For each endpoint with a linked SOA benchmark, the system displays:

- Your device result vs. SOA pooled benchmark vs. similar device results (from SOA extraction grids)
- Visual: bar chart or table
- Auto-feeds the SOA Device comparison table (Module B, Section 9) — the "Your Device" column is now populated
- Quality context: the SOA benchmark quality summary (from QUADAS-2 and Reading Grid assessments) is displayed alongside, so the user can see the evidence quality behind the benchmark

6.4 GSPR Mapping (MDR Annex I Compliance)

6.4.1 Purpose

MDR requires that clinical evaluation confirms conformity with the relevant General Safety and Performance Requirements (GSPRs) from Annex I. Each validation result must be traceable to the GSPR(s) it addresses. Notified Bodies check this mapping as part of the CEAR (Clinical Evaluation Assessment Report).

6.4.2 GSPR Mapping Matrix

The system maintains a mapping between endpoints/results and GSPRs:

GSPR	Requirement Summary	Addressed By	Evidence	Status
Annex I, §1	Devices shall achieve performance intended by manufacturer	Primary endpoint: Sensitivity ≥ 60%, Specificity ≥ 89%	Standalone validation: Sens 90.3%, Spec 91.9%	✅ Confirmed
Annex I, §3	Devices shall be safe and effective for intended users	MRMC: readers perform better with AI assistance	MRMC: IT reduced by 35.7%, $p < 0.0001$	✅ Confirmed
Annex I, §8	Device performance shall not be adversely affected by foreseeable conditions	Stratified analysis across scanner makers, age, sex	Stratified results: no significant performance degradation	✅ Confirmed
Annex I, §14.1	Software shall be designed to ensure repeatability, reliability, performance	Processing time assessment	Secondary endpoint: mean processing time [X]s	✅ Confirmed
Annex I, §14.2	Software with diagnostic purpose: adequate sensitivity/specificity	Primary endpoint performance goals	Standalone: Sens 90.3%, Spec 91.9%	✅ Confirmed
Annex I, §23.4	Clinical benefit demonstrated	MRMC clinical benefit study	Clinical benefit report: reading time reduction demonstrated	✅ Confirmed
...	...	...	...	...

6.4.3 Implementation

- Default GSPR list for SaMD Class IIa/IIb is pre-loaded (configurable)
- When defining endpoints, the user tags which GSPRs the endpoint addresses
- After results are imported, the GSPR matrix is auto-populated with the corresponding evidence
- Unaddressed GSPRs are flagged: "⚠️ GSPR §X has no linked clinical evidence — justify or add evidence"
- The GSPR matrix is exportable as a DOCX/XLSX table for CER inclusion
- The matrix feeds directly into the CER module's GSPR compliance section

6.5 Benefit Quantification (MDR Annex I, Benefit-Risk Analysis)

6.5.1 Purpose

MDR requires a structured benefit-risk determination. Clinical benefits must be quantified — not just "statistically significant" but assigned a Benefit Level (BL) based on probability and magnitude. This feeds the CER's Section 6 (Criteria for Acceptability of Benefit/Risk Ratio).

6.5.2 Benefit Level Calculation

For each demonstrated clinical benefit (from MRMC or standalone results):

Step 1 — Define the benefit:

Field	Example
benefit_id	B1
benefit_description	CINA-CSpine improves radiologists' read time compared to unaided conventional read
source_study	MRMC Clinical Benefit Study
source_endpoint	Primary: Interpretation Time (Aided vs. Unaided)

Step 2 — Quantify probability:

Probability Category	Criteria	Score
Very common	Demonstrated in ≥ 80% of cases/readers	5
Common	Demonstrated in 50-79% of cases/readers	4
Occasional	Demonstrated in 20-49% of cases/readers	3
Uncommon	Demonstrated in 5-19% of cases/readers	2
Rare	Demonstrated in < 5% of cases/readers	1

Step 3 — Quantify magnitude:

Magnitude Category	Criteria	Score
Major	Prevents serious clinical harm or dramatically improves outcomes (e.g., >50% improvement)	5
Significant	Meaningful clinical improvement (e.g., 25-50% improvement)	4
Moderate	Noticeable improvement (e.g., 10-25% improvement)	3
Minor	Small but measurable improvement (e.g., 5-10%)	2
Marginal	Detectable but minimal improvement (< 5%)	1

Step 4 — Benefit Level:

BL = Probability Score × Magnitude Score

Example (CSpine):

> Benefit B1: Reading time reduction of 35.7% (magnitude: Significant = 4)
> Demonstrated across all 4 readers, $p < 0.0001$ (probability: Very common = 5)
> Benefit Level = 5 × 4 = 20 (High)

6.5.3 Benefit Registry

All quantified benefits are stored in a benefit registry per project:

Field	Description
benefit_id	B1, B2, ...
description	Benefit statement
probability_score	1-5 with justification
magnitude_score	1-5 with justification
benefit_level	Calculated BL
source_study_id	Validation study providing the evidence
source_endpoint_id	Specific endpoint result
supporting_data	Key statistics (e.g., "35.7% reduction, $p < 0.0001$")

This registry feeds directly into the CER's benefit-risk analysis section, where it's compared against the risk registry from the Risk Management module.

6.6 Algorithmic Fairness Assessment (MDCG 2020-6 Compliance)

6.6.1 Purpose

Regulators (FDA, EU Notified Bodies) increasingly expect demonstration that AI/ML devices perform equitably across demographic subgroups. This section formalizes the assessment of potential algorithmic bias.

6.6.2 Assessment Framework

After stratified results are imported, the system performs an automated fairness analysis:

For each primary endpoint, across each demographic stratification dimension:

Comparison	Metric	Threshold	Result
Scanner maker (GE vs. Siemens vs. Philips vs. Canon)	Sensitivity delta between best and worst group	> 15% absolute difference	⚠️ Flag for review
Patient age (18-45 vs. 45-75 vs. ≥75)	Sensitivity delta	> 10%	⚠️ Flag
Patient sex (M vs. F)	Sensitivity delta	> 10%	⚠️ Flag
Clinical source (US vs. OUS)	Sensitivity delta	> 10%	⚠️ Flag
Reader expertise (expert vs. non-expert) — MRMC	Accuracy delta	> 10%	⚠️ Flag

Thresholds are configurable per project (some devices have legitimate performance differences across subgroups).

6.6.3 Fairness Report Section

Auto-generated for inclusion in the Validation Report:

- Table: performance metrics per subgroup for all stratification dimensions
- Flagged disparities with magnitude and clinical significance assessment
- Justification for acceptability (required for flagged disparities)
- Overall conclusion: "No clinically significant algorithmic bias was detected across demographic subgroups" OR "Performance disparities were identified in [subgroup X] and are documented as [accepted limitation / PMCF action item]"

6.6.4 Link to PMS

Flagged disparities that cannot be fully resolved are automatically registered as PMS gaps (feeds PMS module) with a PMCF action: "Collect additional validation data on [underperforming subgroup]."

6.7 Labeling Validation (FDA 21 CFR §892.2060 Compliance)

6.7.1 Purpose

FDA special controls require validation that labeling accurately reflects device performance. This assessment verifies alignment between the validated intended use, warnings, and performance claims.

6.7.2 Assessment Checklist

Labeling Element	Validation Check	Source	Status
Intended Use Statement	Does the validated performance support the stated intended use?	Protocol intended use vs. validation results	✅ / ❌
Indications for Use	Are all stated indications supported by validation data?	Protocol scope vs. result coverage	✅ / ❌
Performance Claims	Do claimed performance figures match validation results (within CIs)?	SOA claims vs. validation results	✅ / ❌
Contraindications	Are known failure modes reflected in contraindications?	False positive/negative analysis	✅ / ❌
Warnings	Are validated limitations reflected in warnings? (e.g., "not validated for pediatric use")	Stratified results, known limitations	✅ / ❌
Precautions	Are use conditions reflected? (e.g., scanner requirements, slice thickness)	Dataset description, stratified results	✅ / ❌
User training requirements	Is required training level documented? (based on MRMC reader training protocol)	MRMC training protocol, reader qualification	✅ / ❌

6.7.3 Implementation

- The checklist is populated semi-automatically: labeling elements are imported from the project's labeling documentation, and the system links them to relevant validation results
- Gaps are flagged: "Warning X states [Y], but validation data does not cover [Z]"
- The completed checklist is included as an appendix to the Validation Report
- This is particularly relevant for 510(k) and De Novo submissions

---

7. Report Generation

7.1 Report Types & Assembly

Reports can be generated as separate documents or combined, depending on the regulatory strategy:

Format	Use Case	Example
Separate Standalone Report	Standalone validation as its own appendix	CSpine: Appendix B1
Separate MRMC Report	Clinical benefit as its own appendix	CSpine: Appendix B2
Combined Report	Standalone + MRMC in one document	ASPECTS: single validation report

The user chooses the assembly format when generating the report. The underlying data and sections are the same — only the document assembly differs.

Standalone Validation Report — structure matching your current template:

```
1. Introduction
  ├── 1.1 Document Overview
  ├── 1.2 Abbreviations
  ├── 1.3 Project References
  └── 1.4 Intended Use

2. Methodology
  ├── 2.1 Material
  ├── 2.2 Evaluated Data
  │   └── 2.2.1 Data Collection Process
  ├── 2.3 Data Assessment
  │   ├── 2.3.1 Ground Truth Process
  │   ├── 2.3.2 Final Consensus Validation Database
  │   └── 2.3.3 Post-Processing & Results Reporting
  └── 2.4 Stratification Plan (additional assessments if applicable)

3. Statistical Analysis
  ├── 3.1 Performance Analysis — Primary & Secondary Endpoints
  │   ├── 3.1.1 Primary Endpoint (definition)
  │   ├── 3.1.2 Sample Size & Targeted Performance Estimation
  │   ├── 3.1.3 Additional Supportive Statistics
  │   └── 3.1.4 Secondary Endpoint
  └── 3.2 Additional Assessments

4. Results
  ├── 4.1 Standalone Performance
  │   ├── 4.1.1 Data Distribution
  │   │   ├── Scanner makers (table)
  │   │   ├── Patient age (table)
  │   │   ├── Patient sex (table)
  │   │   └── [Other stratification dimensions] (tables)
  │   ├── 4.1.2 Statistical Results
  │   │   ├── 4.1.2.1 Primary Endpoint Results (confusion matrix + sensitivity/specificity)
  │   │   ├── 4.1.2.2 Secondary Endpoint Results
  │   │   └── 4.1.2.3 Additional Supportive Statistics
  │   └── 4.1.3 Conclusion (per section)
  └── 4.2 Additional Assessments (if applicable)

5. Discussion and Conclusions
  ├── 5.1 Primary Endpoint
  ├── 5.2 Secondary Endpoint
  ├── 5.3 Additional Statistics
  ├── 5.4 Additional Assessments
  ├── 5.5 Comparison to SOA & Similar Device
  ├── 5.6 Algorithmic Fairness Assessment
  └── 5.7 Labeling Validation Summary
```

6. Conclusion

References

Appendices

```
├── Appendix: GSPR Mapping Matrix
├── Appendix: Benefit Quantification Summary
├── Appendix: Algorithmic Fairness Detail Tables
└── Appendix: Labeling Validation Checklist
```

MRMC Clinical Benefit Report — structure matching your current template:

```
1. Introduction
  ├── 1.1 Document Overview
  ├── 1.2 Abbreviations
  └── 1.3 Project References

2. Clinical Benefit Overview
  └── Benefit claims (linked to SOA similar device analysis)

3. Clinical Benefit Study
  ├── 3.1 Methodology
  │  ├── 3.1.1 General Study Design
  │  ├── 3.1.2 Evaluated Data
  │  ├── 3.1.3 Readers' Qualification (table)
  │  └── 3.1.4 Reader Training
  ├── 3.2 Statistical Analysis
  │  ├── 3.2.1 Benefit Claim
  │  ├── 3.2.2 Primary Endpoint
  │  ├── 3.2.3 Additional Supportive Assessments
  │  │  ├── a) Stratified Statistics
  │  │  └── b) Additional Statistics
  │  └── 3.2.4 Secondary Endpoint
  └── 3.3 Results
     ├── 3.3.1 Final Cohort
     ├── 3.3.2 Data Distribution
     ├── 3.3.3 Primary Endpoint Results
     ├── 3.3.4 Additional Supportive Assessments
     │  ├── 3.3.4.1 Stratified Analyses Results
     │  └── 3.3.4.2 Additional Analyses Results
     └── 3.3.5 Secondary Endpoint Results

4. Discussion

5. Conclusion

6. Bibliography
```

Combined Standalone + MRMC Report (ASPECTS pattern):

```
1. Introduction
  ├── 1.1 Document Overview
  ├── 1.2 Project References
  └── 1.3 Intended Use

2. [Device] Pivotal Standalone Performance Testing and Clinical MRMC Studies
  ├── 2.1 Methodology
  │  ├── 2.1.1 Material
  │  ├── 2.1.2 Evaluated Data
  │  │  ├── Data Collection
  │  │  └── Data Distribution
  │  ├── 2.1.3 Data Assessment
  │  │  ├── Ground Truth Establishment
  │  │  ├── Standalone Performance Evaluation Design
  │  │  └── Clinical Performance Evaluation Design (MRMC)
  │  │     ├── Study Design (cross-over, randomization, washout)
  │  │     ├── Readers' Qualification (table)
  │  │     ├── Reader Training
  │  │     ├── Rating Scale Definition
  │  │     └── Study Arms & Blinding
  │
  ├── 2.2 Statistical Analysis
  │  ├── 2.2.1 Pivotal Standalone Performance Testing
  │  │  ├── Primary Endpoint
  │  │  ├── Secondary Endpoint
  │  │  └── Sample Size & Targeted Performance Estimation
  │  └── 2.2.2 Clinical MRMC Study
  │     ├── Primary Endpoint (accuracy, paired comparison)
  │     ├── Secondary Endpoint (reading time)
  │     └── Additional Assessments (ROC AUC, stratified, score-based)
  │
  ├── 2.3 Results
  │  ├── 2.3.1 Data Distribution (shared dataset)
  │  ├── 2.3.2 Pivotal Standalone Performance Testing Results
  │  │  ├── Primary Endpoint Results
  │  │  ├── Secondary Endpoint Results
  │  │  ├── Additional Assessments Results
  │  │  └── Conclusion
  │  └── 2.3.3 Clinical MRMC Study Results
  │     ├── Primary Endpoint Results
  │     ├── Stratified Results (by reader, expertise, region, scanner...)
  │     ├── Secondary Endpoint Results (reading time)
  │     ├── Additional Assessments (ROC AUC, region-based, score-based)
  │     └── Conclusion
  │
  └── 2.4 Summary and Discussion
```

```
    ├── Standalone Performance Testing Discussion
    └── MRMC Study Discussion

3. Conclusion

4. Bibliography

Appendices
```

7.2 Generation Process

1. AI drafts each section based on:
    - Protocol data (methodology, endpoints, dataset description)
    - Imported results (data distribution tables, statistical results)
    - SOA context (benchmarks, similar device data, quality assessment summaries)
    - Comparison analysis (pass/fail, justifications)
2. Tables are auto-populated:
    - Data distribution tables → directly from imported XLS
    - Confusion matrices → from imported data
    - Results tables → from endpoint mapping
    - Reader qualification table → from reader records (MRMC)
3. User reviews and edits each section
4. Export as DOCX with proper formatting, tables, and cross-references

7.3 Auto-Generated Content

The system auto-generates specific content blocks:

- **Sample size justification paragraph** — from calculator inputs
- **Confusion matrix table** — from imported data
- **Data distribution tables** — from imported XLS
- **Primary endpoint results paragraph** — e.g., "The device achieved a sensitivity of 90.3% [84.5–94.5%], exceeding the performance goal of ≥ 60%. The specificity was 91.9% [86.8–95.5%], exceeding the performance goal of ≥ 89%."
- **SOA comparison paragraph** — e.g., "Compared to the similar device BriefCase (Aidoc Medical), whose sensitivity ranged from 54.9% to 91.7%, CINA-CSpine's sensitivity of 90.3% places it at the upper end of the SOA performance range."
- **Evidence quality context** — auto-generated summary referencing the SOA's quality assessments: "The SOA benchmarks are derived from N pivotal studies, of which X were rated high-quality (QUADAS-2: low risk of bias) and Y were rated 'Very good' or 'Pretty good' on internal methodological assessment."

- **Discussion points per endpoint** — AI drafts based on result vs. benchmark analysis
- **Abbreviations table** — auto-compiled from all abbreviations used in the document

---

8. Patch Validation (Lightweight Re-validation)

8.1 Concept

When a software patch is released (bug fix, minor algorithm update), a full re-validation may not be required. A patch validation is a lighter process:

- Created as `study_type = 'standalone_patch'` with `parent_study_id` pointing to the original standalone study
- Inherits the protocol from the parent study (endpoints, acceptance criteria, methodology)
- The database may change (new cases added, some excluded) — dataset description is updated
- Results are re-imported and re-evaluated against the same acceptance criteria
- The report is shorter: focuses on what changed (dataset delta, result delta) and confirms performance is maintained

8.2 Patch Report Structure

```
1. Introduction (abbreviated — references parent study)

2. Changes from Parent Study
   ├── Software version delta
   ├── Dataset changes (added/removed cases, new sources)
   └── Methodology changes (if any — with justification)

3. Results
   ├── Data Distribution (updated)
   ├── Statistical Results (updated)
   └── Comparison to Parent Study Results (delta table)

4. Conclusion
```

8.3 Delta Table

Endpoint	Parent Study (v1.0)	Patch (v1.1)	Delta	Status
Sensitivity	90.3% [84.5–94.5%]	91.1% [85.4–95.1%]	+0.8%	✅ Maintained
Specificity	91.9% [86.8–95.5%]	90.8% [85.6–94.7%]	-1.1%	✅ Maintained
...	...	...	...	...

9. Study Workflow & Statuses

9.1 Lifecycle

draft → protocol_approved → data_collection → analysis → report_draft → completed → locked

Status	What Happens	What's Editable
draft	Protocol being written: endpoints, dataset plan, readers, sample size	Everything
protocol_approved	Protocol reviewed and approved — testing can begin	Protocol: warning on changes (see 9.2)
data_collection	Data being collected by clinical sources, GT process ongoing	Dataset description updates
analysis	Data Science running analysis, XLS being imported	Data imports (versioned)
report_draft	Report being generated and reviewed	Report content, justifications
completed	Report finalized, ready for CER	Minor edits (logged)
locked	Frozen for a CER version. Read-only.	Nothing

9.2 Protocol Amendment Handling

Once a protocol is approved, changes are allowed but warned:

- User modifies an endpoint, acceptance criterion, or methodology
- System shows: ⚠️ "This protocol has been approved. Modifying it after approval will be logged in the audit trail."
- The change is saved with:
  - Timestamp
  - What changed (old value → new value)
  - Who changed it
  - No justification required (soft process per your request)
- The protocol version is incremented automatically (1.0 → 1.1)
- Amendment history is visible in the protocol and included in the report

10. Exports & Outputs

Export	Format	Content
Study Summary / Protocol	DOCX	Full protocol document with all sections
Validation Report	DOCX	Complete report (standalone or MRMC structure)
Patch Validation Report	DOCX	Abbreviated report with delta analysis
Results Summary	XLSX	Endpoint results, comparisons, pass/fail
Data Distribution	XLSX	All stratification tables
GSPR Mapping Matrix	XLSX / DOCX	GSPRs × evidence mapping with status
Benefit Quantification	XLSX / DOCX	Benefit registry with BL calculations
Algorithmic Fairness Report	XLSX / DOCX	Subgroup performance comparison, flagged disparities
Labeling Validation Checklist	XLSX / DOCX	Labeling elements × validation evidence
Import Template	XLSX	Empty standardized template for Data Science team

11. Async Operations

Operation	Estimated Duration	Async?
XLS import & parsing	2-10s	No (inline with spinner)
Report generation (full)	30-90s	Yes
Section drafting (single)	10-30s	No (inline)
Version comparison	2-5s	No (inline)

12. Permissions

Role	Can Do
clinical / regulatory	Full access: create studies, write protocols, import data, generate reports

Role	Can Do
manager	Full access + approve protocols, lock/unlock studies
collaborateur	Read-only access to completed/locked studies and exports
data_science	Upload XLS data imports (limited access — no protocol editing)
admin	Everything + manage import templates at system level

13. Open Items

1. **Sample size calculator — internal computation:** The calculator implements the Hajian-Tilaki method (Clopper-Pearson CI for one-sample sensitivity/specificity) internally. This covers the core use case. For advanced methods (Obuchowski-Rockette power analysis for MRMC), the system provides the AI-assisted recommendation and the user can validate with external tools if needed.
2. **Reverse calculator — statistical method library:** The system needs a library of common statistical methods with their sample size formulas, assumptions, and trade-offs. Initial library: Clopper-Pearson exact binomial, Wilson score interval, Bootstrap CI, Fisher's exact test, paired t-test, McNemar's test. This library is extensible.
3. **OR-ANOVA implementation:** The Obuchowski-Rockette ANOVA + Jackknife method (used for ASPECTS ROC AUC analysis) is implemented in R (MRMCaov package). CORTEX imports the results from the Data Science team's R output — we don't reimplement the analysis itself, but we must correctly parse and display the OR-ANOVA results (AUC estimates, p-values, CIs per reader and overall).
4. **Rating scale flexibility:** MRMC studies may use binary (CSpine) or ordinal scales (ASPECTS: 6-point). The data import and results display must handle both. The endpoint definition captures the rating scale type and cutoff point for binary conversion.
5. **Discussion section AI quality:** Auto-generating Discussion remains the hardest part. Structured prompts help: "Key findings?" → "Comparison to SOA?" → "Comparison to similar device?" → "Limitations?" → "Conclusion?" The AI drafts, the user heavily edits. Quality will improve with fine-tuning on your existing reports. The addition of SOA quality assessment context (evidence quality summary) improves the AI's ability to draft contextually accurate discussion points.
6. **Combined vs. separate report logic:** The system needs a flexible document assembly engine that can combine standalone + MRMC sections into one document or keep them separate, reusing the same underlying data. This engine will also serve the CER module later.
7. **Mini-SLS consistency:** The mini-SLS for MRMC methodology (Section 4.6) uses the full SLS module capabilities including exclusion codes and live PRISMA. However, the PRISMA flowchart export for a mini-SLS may use a simplified layout (fewer queries, fewer databases) — the system should auto-adapt the flowchart complexity to the search scope.